Jovan Komatovic

Senior Researcher @ Category Labs

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Profiles: [DBLP] [Google Scholar]

Research interests

Designing robust and efficient distributed algorithms and obtaining new impossibility results and lower bounds in the realm of fault-tolerant distributed computing

Keywords: Distributed computing, Byzantine fault tolerance, Consensus

Education

2019 - 2025 **PhD in Computer Science**, *École Polytechnique Fédérale de Lausanne (EPFL)*, Lausanne, Switzerland
- Research focus: Byzantine fault-tolerant distributed computing
- Supervisor: Rachid Guerraoui
- Thesis title: Optimal Byzantine Agreement with Little Cryptography
- *Principles of Distributed Computing Doctoral Dissertation Award (PODC/DISC 2026)*

2018 - 2019 **MSc in Applied Mathematics**, *University of Belgrade*, Belgrade, Serbia
- GPA: 9.83/10
- Thesis: Providing Accountability Support for the Tendermint Consensus Algorithm

2014 - 2018 **BSc in Software Engineering**, *University of Belgrade*, Belgrade, Serbia
- GPA: 9.84/10

Experience

2025 - present **Senior Researcher**, *Category Labs*, New York, USA

Summer 2024 **Research Intern**, *a16z crypto*, New York, USA
Supervised by Professor Tim Roughgarden

Honours & Awards

PODC/DISC 2026 **Principles of Distributed Computing Doctoral Dissertation Award**
Awarded jointly by ACM PODC and EATCS DISC for the thesis "Optimal Byzantine Agreement with Little Cryptography"

DISC 2023 **Best Paper Award**
Awarded for the paper "Every Bit Counts in Consensus"

DISC 2022 **Best Student Paper Award**
Awarded for the paper "Byzantine Consensus is $\Theta(n^2)$: The Dolev-Reischuk Bound is Tight even in Partial Synchrony!"

ICDCS 2022 **Best Paper Award**
Awarded for the paper "Crime and Punishment in Distributed Byzantine Decision Tasks"

IPDPS 2022 **Best Paper Award**
Awarded for the paper "As easy as ABC: Optimal (A)ccountable (B)yzantine (C)onsensus is easy!"

DSN 2020 **Runner-Up for Best Paper Award**
Awarded for the paper "Online Payments by Merely Broadcasting Messages"

EPFL 2023 **Teaching Assistance Excellence Award**
Awarded for teaching excellence

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Publications [Author names in alphabetical order]

Peer-Reviewed Conference Publications

DISC 2026 **AegisBFT: Fast, Responsive, Fork-Resistant Consensus with Speculation Accountability**
M. M. Jalalzai, K. Babel, J. Komatovic, T. Klenze, S. Das, F. Elsheimy, M. Setrin, J. Bergschneider, B. P. Gilkalaye
International Symposium on Distributed Computing
[Author names not in alphabetical order]

PODC 2026 **Forget-IT: Optimal Good-Case Latency for Information-Theoretic BFT**
I. Abraham, S. Das, Y. Efron, J. Komatovic
ACM Symposium on Principles of Distributed Computing
Invited to Special Edition of Distributed Computing (DC)

ICDCS 2026 **Hash-Based Asynchronous MVBA with Optimal Complexity and Near-Optimal Resilience**
J. Komatovic, J. Neu, T. Roughgarden
IEEE International Conference on Distributed Computing Systems

S&P 2026 **Scalable Accountable Byzantine Agreement and Beyond**
P. Civit, D. Collins, V. Gramoli, R. Guerraoui, J. Komatovic, M. Vidigueira, P. Zarbafian
IEEE Symposium on Security and Privacy

DISC 2025 **Byzantine Consensus in the Random Asynchronous Model**
G. Danezis, J. Komatovic, L. Kokoris-Kogias, A. Sonnino, I. Zablotchi
International Symposium on Distributed Computing

AFT 2025 **From Permissioned to Proof-of-Stake Consensus**
J. Komatovic, A. Lewis-Pye, J. Neu, T. Roughgarden, E. N. Tas
Advances in Financial Technologies

PODC 2025 **Repeated Agreement is Cheap! On Weak Accountability and Multishot Byzantine Agreement**
P. Civit, M. A. Dzulfikar, S. Gilbert, R. Guerraoui, J. Komatovic, M. Vidigueira
ACM Symposium on Principles of Distributed Computing

Eurocrypt 2025 **Juggernaut: Efficient Crypto-Agnostic Byzantine Agreement**
D. Collins, Y. Efron, J. Komatovic
International Conference on the Theory and Applications of Cryptographic Techniques

SODA 2025 **Partial Synchrony for Free: New Upper Bounds for Byzantine Agreement**
P. Civit, M. A. Dzulfikar, S. Gilbert, R. Guerraoui, J. Komatovic, M. Vidigueira, I. Zablotchi
ACM-SIAM Symposium on Discrete Algorithms

DISC 2024 **Efficient Signature-Free Validated Agreement**
P. Civit, M. A. Dzulfikar, S. Gilbert, R. Guerraoui, J. Komatovic, M. Vidigueira, I. Zablotchi
International Symposium on Distributed Computing

PODC 2024 **All Byzantine Agreement Problems are Expensive**
P. Civit, S. Gilbert, R. Guerraoui, J. Komatovic, A. Paramonov, M. Vidigueira
ACM Symposium on Principles of Distributed Computing

PODC 2024 **DARE to Agree: Byzantine Agreement with Optimal Resilience and Adaptive Communication**
P. Civit, M. A. Dzulfikar, S. Gilbert, R. Guerraoui, J. Komatovic, M. Vidigueira, I. Zablotchi
ACM Symposium on Principles of Distributed Computing

DISC 2023 **Every Bit Counts in Consensus**
P. Civit, S. Gilbert, R. Guerraoui, J. Komatovic, M. Monti, M. Vidigueira
International Symposium on Distributed Computing
Best Paper Award
Invited to Journal of the ACM (JACM)

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PODC 2023 **On the Validity of Consensus**
P. Civit, S. Gilbert, R. Guerraoui, J. Komatovic, M. Vidigueira
ACM Symposium on Principles of Distributed Computing

DISC 2022 **Byzantine Consensus is $\Theta(n^2)$: The Dolev-Reischuk Bound is Tight even in Partial Synchrony!**
P. Civit, M. A. Dzulfikar, S. Gilbert, V. Gramoli, R. Guerraoui, J. Komatovic, M. Vidigueira
International Symposium on Distributed Computing
Best Student Paper Award
Invited to Special Edition of Distributed Computing (DC)

ICDCS 2022 **Crime and Punishment in Distributed Byzantine Decision Tasks**
P. Civit, S. Gilbert, V. Gramoli, R. Guerraoui, J. Komatovic, Z. Milosevic, D.-A. Seredinschi
IEEE International Conference on Distributed Computing Systems
Best Paper Award

IPDPS 2022 **As easy as ABC: Optimal (A)ccountable (B)yzantine (C)onsensus is easy!**
P. Civit, S. Gilbert, V. Gramoli, R. Guerraoui, J. Komatovic
IEEE International Parallel & Distributed Processing Symposium
Best Paper Award
Invited to Special Edition of Journal of Parallel and Distributed Computing (JPDC)

OPODIS 2020 **Dynamic Byzantine Reliable Broadcast**
R. Guerraoui, J. Komatovic, P. Kuznetsov, Y.-A. Pignolet, D.-A. Seredinschi, A. Tonkikh
International Conference on Principles of Distributed Systems

DSN 2020 **Online Payments by Merely Broadcasting Messages**
D. Collins, R. Guerraoui, J. Komatovic, P. Kuznetsov, M. Monti, M. Pavlovic, Y.-A. Pignolet, D.-A. Seredinschi, A. Tonkikh, A. Xygkis
IEEE/IFIP International Conference on Dependable Systems and Networks
Runner-Up for Best Paper Award

Short Peer-Reviewed Conference Publications

DSN-S 2022 **Revisiting Tendermint: Design Tradeoffs, Accountability, and Practical Use**
E. Buchman, R. Guerraoui, J. Komatovic, Z. Milosevic, D.-A. Seredinschi, J. Widder
IEEE/IFIP International Conference on Dependable Systems and Networks (Supplements)

Peer-Reviewed Journal Publications

TDSC 2025 **Carbon: Scaling Trusted Payments with Untrusted Machines**
M. Camaioni, R. Guerraoui, J. Komatovic, M. Monti, P.-L. Roman, M. Vidigueira, G. Voron
IEEE Transactions on Dependable and Secure Computing

DC 2023 **Byzantine Consensus is $\Theta(n^2)$: The Dolev-Reischuk Bound is Tight even in Partial Synchrony!**
P. Civit, M. A. Dzulfikar, S. Gilbert, V. Gramoli, R. Guerraoui, J. Komatovic, M. Vidigueira
Distributed Computing
Invited

JPDC 2023 **As easy as ABC: Optimal (A)ccountable (B)yzantine (C)onsensus is easy!**
P. Civit, S. Gilbert, V. Gramoli, R. Guerraoui, J. Komatovic
Journal of Parallel and Distributed Computing
Invited

Preprints

arXiv 2026 **Quantum Cadence: Post-Quantum Consensus with Multiple Concurrent Proposers**
M. M. Jalalzai, T. Klenze, J. Komatovic

ePrint 2026 **Byzantine Broadcast in $\widetilde{O}(n^2)$ Communication**
Y. Efron, J. Komatovic, L. Ren

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ePrint 2026 **Simple Byzantine Lattice Agreement in $O(\frac{\log f}{\log \log f})$ Rounds**
Y. Efron, J. Komatovic

ePrint 2026 **Floor-IT: Information-Theoretic BFT in Partial Synchrony with Two Round Good Case Latency and Optimal Resilience**
I. Abraham, Y. Efron, J. Komatovic, A. Ranchal-Pedrosa

arXiv 2026 **Cadence: Extreme Pipelining with Multiple Concurrent Proposers**
K. Babel, F. Elsheimy, L. Heimbach, M. M. Jalalzai, T. Klenze, J. Komatovic, J. Milionis, M. Setrin, V. Shoup

ePrint 2026 **Computing in a Safe House: Accountable Universally Composable Asynchronous Secure Distributed Computing**
P. Civit, D. Collins, V. Gramoli, R. Guerraoui, J. Komatovic, M. Vidigueira, P. Zarbafian

arXiv 2025 **MonadBFT: Fast, Responsive, Fork-Resistant Streamlined Consensus**
M. M. Jalalzai, K. Babel, J. Komatovic, T. Klenze, S. Das, F. Elsheimy, M. Setrin, J. Bergschneider, B. Gilkalaye
[Author names not in alphabetical order]

arXiv 2023 **Strong Byzantine Agreement with Adaptive Word Complexity**
P. Civit, S. Gilbert, R. Guerraoui, J. Komatovic, M. Vidigueira

Student Mentoring

Fatima Elsheimy, *Research Intern*, Category Labs
Javier Nieto, *Research Intern*, Category Labs
Emmanouil Chatzakis, *MSc Computer Science*, EPFL
Othmane Sqalli Houssaini, *MSc Computer Science*, EPFL
Rui Huang, *MSc Computer Science*, EPFL
Marija Katanic, *MSc Computer Science*, EPFL
Anton Paramonov, *MSc Computer Science*, EPFL
Stella Petronio, *MSc Computer Science*, EPFL
Andre Piccione, *MSc Computer Science*, EPFL
Yang Yu-Hsin, *MSc Computer Science*, EPFL
Rufer Maxime Valentin, *BSc Computer Science*, EPFL

Reviewing Activities

PODC ACM Symposium on Principles of Distributed Computing
DISC International Symposium on Distributed Computing
SODA ACM-SIAM Symposium on Discrete Algorithms
OPODIS International Conference on Principles of Distributed Systems
CRYPTO Annual International Cryptology Conference
TCC Theory of Cryptography Conference
TDSC Transactions on Dependable and Secure Computing
JACM Journal of the ACM
SRDS IEEE International Symposium on Reliable Distributed Systems
GandALF International Symposium on Games, Automata, Logics, and Formal Verification

Grants

2021 **Interchain Foundation**, 90'000 USD
Project: Cross-chain validation: Reusing trust in multi-chained systems
Co-written with PI Rachid Guerraoui (EPFL)

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Teaching

Fall 2020 - 2024 **Distributed Algorithms**, *MSc Computer Science*, EPFL
Teaching assistant

Spring 2020 **Software Development Project**, *BSc Computer Science*, EPFL
Teaching assistant

Spring 2021 **Practice of Object-Oriented Programming**, *BSc Computer Science*, EPFL
Teaching assistant

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